Chapter 1

Introduction

1.1 Motivation

It is always said that we are in the information age. As a witnesses of this information age, maybe the most observable evidence we have is the rapid proliferation of multimedia data, such as speech, audio, images, video, animation and graphics. As more and more means are provided to produce digital media information, the multimedia information not only increase, it accelerates. As an example, according to the magazine of *Advanced Imaging*, over 2700 photographs are taken every second in the year of 2000, adding up to 80 billion new images. Approximately 2 billion chest x-rays, mammograms, CT scans and other x-rays are taken every year as well. Further, with the rise in popularity of consumer digital imaging and of more film photographers using Web photo services, a significant portion of these images are stored online to be accessed via the internet. For example, Photo Works alone already hosts over 100 million images on its Web site. This rapid proliferation poses many challenges in information precessing. Clearly, efficient tools for storage, manipulation and communication are urgently needed to ingest and utilize such large amount of multimedia data.

Since the early 90's, image/video information retrieval has emerged as an active research area. Multimedia information search and retrieval has applications in digital library, education, image/video indexing, publishing and editing and on-line shopping. In the last ten years, a lot of effort have been dedicated to develop the techniques involved in image/video information retrieval. The IBM's QBIC project [27] played an important role in the early 90's in inspiring effort along this direction. Some of the main players include Virage (http://

www.virage.com), Context Media (http://www.contextmedia.com), Covera (http://www.convera.com, previously called Excalibur). In academia, the effort can be highlighted by some prototype systems developed in recent years, such as Netra from the University of California, Santa Barbara, VisualSEEK and WebSEEK from th Columbia University, MARS from the University of Illinois at Urbana-Champain, Blobworld from the University of California, Berkeley, PicHunter from MIT and MultimediaMiner from the Simon Fraser University (Canada). The prior work addressed one or more of the following issues:

- image/video indexing: Indexing in content based image/video retrieval usually refers search using one or more visual features. Most of the above mentioned work address this issue. The focus is to provide visual cues that can be ingested by a database and indexed. For example, Virage's VideoLogger$^{(TM)}$ can facilitate key frame detection, shot detection, face detection/recognition, speech ID detection and optical character recognition. The detected information is ingested by database so that user can use the cues, summarized from the original digital resources, to access the image/video information according to their interests.
- similarity search: Similarity search is often distinguished from nearest neighbor search. Often the nearest neighbors of a query in the feature space do not result in or capture what the user is interested in searching the image/video database. In recent years, there has been extensive work on designing techniques that take into account the user's feedback in computing better matches. This involves both the design of sophisticated interfaces for the users to provide the feedback and learning algorithms that improve the search results based on such feedback. For example, VisualSEEK provides a sketch board for the users to express the visual concepts. In Blobworld, users can compose a query by using objects segmented from images. Relevance feedback techniques [38], [69], have been widely used in modifying the search results. For example, in MARS [69], the weights associated with the different cues are modified during the learning process. PicHunter [56] uses the user's feedback to estimate the probability of one image to be an interested image.

- image/video mining: A general objective here is to discover hidden patterns or relationships in a given set of image/video data. Research on multimedia data mining is still in its infancy. Potential opportunities include designing better feature descriptors and efficient indexing in high dimensional spaces by exploiting the underlying feature relationships.

Figure 1-1 shows a schematic of a generic image/video management system. The raw image/video data is first processed to extract the various descriptors. Typical image features include color, texture, shape, objects resulting from a segmentation of the image, and spatial layout of the objects. Motion features provide additional information in case of video data. More specific object oriented features include localization of faces and face descriptor for recognition. A given feature may have more than one associated descriptors. For example, color can be described by both a color histogram or a set of dominant colors in the image.

The communication between the user and the feature descriptors occurs through a data management system. Users typically do not have access to the feature extraction process. Also, typical indexing and search methods do not allow any modification to the feature space. With the advent of more sophisticated data mining methods, one may anticipate automated changes to the feature space, or even feature extraction.

The issue of searching for nearest neighbors when the feature dimensionality is high, is an active research area within the database community. Typical feature dimensions may range from few tens to a few hundreds. Traditional database indexing methods do not scale well beyond 10-15 dimensions, and thus pose challenging problems for large media data management. This is often referred as the curse of dimensionality in the database literature. The problem of searching image/video data is further complicated by the fact that even if efficient methods for nearest neighbor search exist, they do not necessarily imply good similarity matches in the context of similarity search. Perceptual similarity does not often translate to feature vectors that are close to each other in a given feature space.

Much of the research emphasis during the last few years has been on the descriptor computation. These descriptors constitute the basic means in accessing the image/video informa-

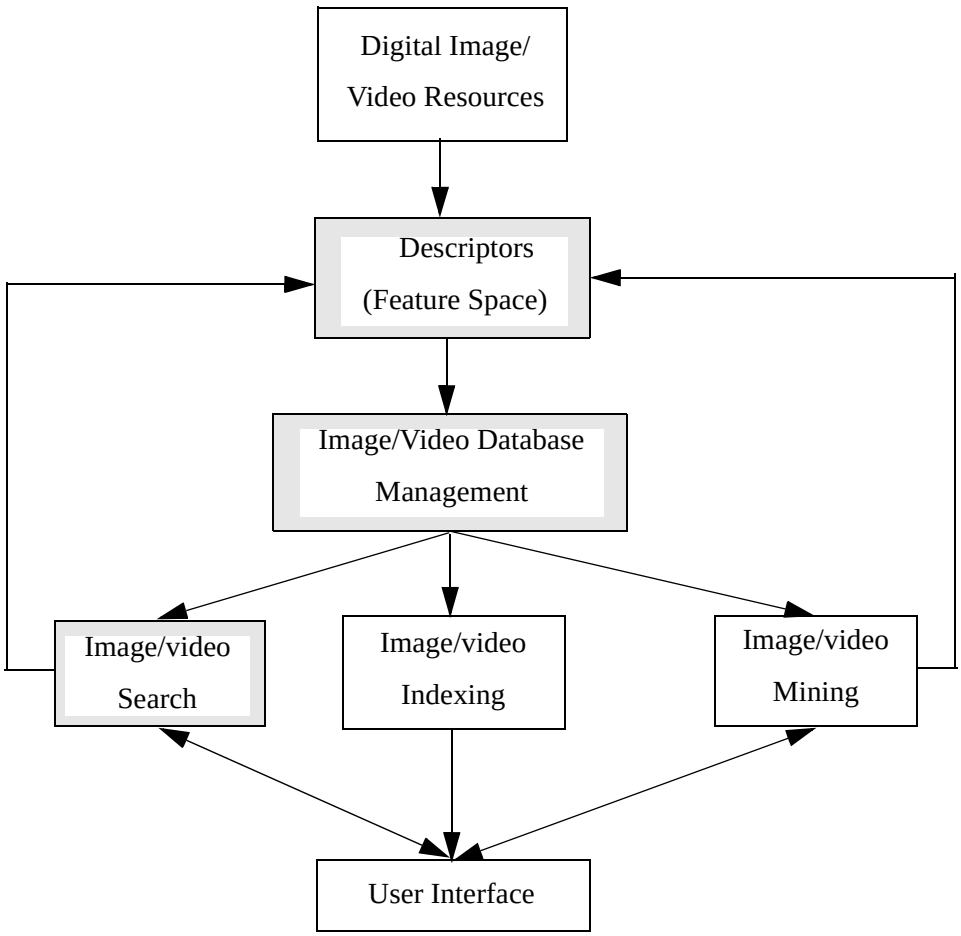

FIGURE 1-1. Schematic of a generic image/video management system

tion. The upcoming international standard, MPEG-7, provides a good showcase of the achievements in this area.

Despite all of the effort and progress made during the last decade, the state of the art content based retrieval lags much behind the general expectations. While its usefulness has been generally acknowledged, the potential applications are yet to be demonstrated. None of the existing database systems support multimedia data beyond keyword/text indexing. The image processing and computer vision research has led to significant progress in feature extraction and data analysis. There has been much progress in high dimensional indexing within the database research. One of the contributions of this thesis is to identify and address some of the challenging issues that are at the interface of visual analysis and database indexing. This includes learning in relevance feedback while dealing with high dimensional features and very large databases.

In developing image/video search techniques, learning algorithms have been widely used in associating user's concept with the descriptors [38], [69]. During the learning process, the algorithm needs to access the descriptors associated with image/video objects. Most work in this area usually ignores the cost to access the high dimensional descriptors through database management system in considering the efficiency of the learning algorithm. On the other hand, the impact of using learning on the indexing structure has generally been ignored. As a result, the learning algorithms are typically evaluated over a very limited number of image/video objects and without considering the cost of accessing descriptors. The feasibility of learning algorithms such developed has not been evaluated for handling large amount of image/video objects.

In summary, it is the emphasis on the synergic design of tools in improving the image/video information retrieval that builds the motivation for this dissertation. Specifically, this emphasis is introduced into developing techniques involved with the three fundamental buildings blocks in Figure 1-1, i.e., the development of the descriptors, the image/video database management system and the image/video search.

1.2 Contributions

There is a clear need for data structures that support the efficient indexing of large amount of objects in high dimensional feature spaces. The research in this area shows that most index structures have an exponential dependence, in space or time or both, upon the number of dimensions. Since the descriptors must be ingested by the database and the application has to go through the database management system to access the descriptors, the synergic design unavoidably needs to consider this difficulty of indexing high dimensional space in developing the descriptors and the search techniques.

Traditionally, the work on developing descriptors focuses the effectiveness of the descriptor in characterizing the image/video objects. However the dependence on the number of dimensions gives rise to the need to consider some other relevant issues in developing and utilizing the descriptors. First, we emphasis the importance of the descriptor's formation. The formation of a descriptor refers to the dimensionality of the descriptor, the data type (float, int, etc.) of the elements and the composition of the elements. Secondly, we consider the dimensionality reduction as an important approach to relieve the burden of indexing the high dimensional space. While considering the search techniques, we not only consider the effectiveness of the techniques in capturing user's presented criteria but also the design of efficient indexing schemes for fast access and retrieval.

The main contributions of this dissertation can be summarized as:

- Development of a compact descriptor for browsing large texture image database. Only 12 bits are needed to represent the 5 dimensional descriptor. Its effectiveness is evaluated over a wide range of texture images.
- Development of an efficient dimensionality reduction algorithm for image retrieval applications.
- Development of new performance measures to explore the performance of dimensionality reduction algorithms.
- Development of an adaptive index structure for nearest neighbor search in high dimensions.
- Development of an adaptive indexing scheme for nearest neighbor search in relevance feedback.

1.3 Dissertation Outline

The organization of this dissertation is as follows:

Chapter 2 provides a brief introduction to the descriptors investigated in this dissertations.

Chapter 3 presents the extraction and evaluation of the texture browsing descriptor.

Chapter 4 presents a dimensionality reduction algorithm for image/video retrieval applicaitons. Meanwhile, two performance evaluation measures are presented to explore the effectiveness of the dimensionality reduction algorithm

Chapter 5 presents an adaptive index structure for similarity search in large image/video databases

Chapter 6 presents an novel index scheme to support the nearest neighbor search in relevance feedback.

Chapter 7 ends the dissertation with a conclusion and a brief discussion of future work.